IoT and Embedded System Based Secure Vault

SUMMER TRAINING REPORT

Submitted in partial fulfilment of the requirements for the award of the degree

of

BACHELOR OF TECHNOLOGY

in

ELECTRONICS & COMMUNICATION ENGINEERING

by

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CANDIDATE'S DECLARATION

It is hereby certified that the work which is being presented in the Summer Training Report entitled “IOT AND EMBEDDED SYSTEM BASED SECURE VAULT” in partial fulfilment of the requirements for the award of the degree of Bachelor of Technology and submitted in the Department of Electronics & Communication Engineering of Bharati Vidyapeeth's College of Engineering, New Delhi (Affiliated to Guru Gobind Singh Indraprastha University, Delhi) is an authentic record of my own work, carried out as the Group Leader of Team “ECE TECH”, during the period from 17th July 2026 to 23rd July 2026 under the guidance of Mr. Armaan Hussain, Uptech Automation.

The matter presented in this Summer Training Report has not been submitted by me for the award of any other degree of this or any other Institute.

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This is to certify that the above statement made by the candidate is correct to the best of my knowledge.

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ABSTRACT

This report presents the design, development, and testing of an IoT and Embedded System Based Secure Vault, built around the ESP32 microcontroller and integrated with the ESP RainMaker cloud platform. The project addresses a clear gap in existing locking mechanisms: traditional key-based lockers depend entirely on a physical key that can be lost, duplicated, or picked, while modern password/keypad-based lockers, although more convenient, still offer no protection against physical tampering and no way to monitor or control the vault remotely.

The prototype combines a 4x4 matrix keypad for PIN-based access, a servo-driven locking mechanism, an ultrasonic sensor for user presence detection, a 16x2 I2C LCD for status display, and an SW-420 vibration sensor for anti-tamper detection. The entire system is coordinated through a non-blocking, state-machine-based firmware architecture running on the ESP32, ensuring that safety-critical checks — vibration monitoring and remote RainMaker commands — are evaluated on every control loop pass regardless of which screen or mode is currently active.

Remote functionality is provided through ESP RainMaker: the vault can be unlocked from a phone application, but only when the user has explicitly selected the Phone/RainMaker access mode on the keypad menu, ensuring that a phone command cannot silently override the vault outside a deliberate access request. A dedicated anti-tamper mechanism differentiates a genuine forceful hit (such as an attempted break-in) from an incidental light touch by sampling the vibration sensor output in short rapid bursts, and responds with an audible buzzer alarm, a rapidly blinking red LED indicator, and a push notification, all of which can only be cleared by an authorised keypad entry or a remote app command.

During development and hardware testing, a recurring brownout reset was diagnosed and traced to the instantaneous current spike drawn by the servo motor at the moment of unlocking. This was resolved by introducing a 3300µF/16V decoupling capacitor across the main power rail, restoring stable operation across repeated unlock/lock cycles. The completed prototype was tested for password-based access, wrong-password lockout, remote phone-based access, and tamper-detection scenarios, with all functions performing as intended.

ACKNOWLEDGEMENTS

I would like to express my sincere gratitude to all those who supported and guided me throughout the course of this summer training and the development of this project.

I am profoundly grateful to our esteemed Head of Department, Mrs. Kusum tharani, for her continuous support and encouragement, which provided a strong academic foundation for this work.

I extend my deepest appreciation to my mentor, Mr. Armaan Hussain of Uptech Automation, for his invaluable technical guidance throughout the training period. His practical insights into embedded systems and IoT integration were instrumental in shaping the design and troubleshooting of this project, particularly in diagnosing and resolving hardware-level issues such as power stability and sensor tuning.

I would also like to extend my special thanks to Mr. Prem Sir and Mr. Amit Shinde Sir for their constant support and valuable help throughout the project.

I am also thankful to my fellow team members of Team “ECE TECH” for their collaboration and support throughout this project.

Finally, I would like to thank my family for their constant encouragement and understanding.

Sincerely,
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LIST OF ABBREVIATIONS

ESP32	Espressif Systems 32-bit Microcontroller
IoT	Internet of Things
LCD	Liquid Crystal Display
I2C	Inter-Integrated Circuit
PIR/IR	Infrared
GPIO	General Purpose Input/Output
PWM	Pulse Width Modulation
BLE	Bluetooth Low Energy
OTA	Over-The-Air (firmware update)
POP	Proof of Possession
V	Volt
A	Ampere
µF	Microfarad
RTOS	Real-Time Operating System

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1. INTRODUCTION

1.1 IoT and Embedded System Based Secure Vault

Security of physical valuables—such as cash, important documents, jewellery, confidential records, and other valuable possessions—has always been an essential requirement in both residential and commercial environments. For centuries, people have relied on some form of locking mechanism to protect these assets from theft, unauthorized access, or accidental misuse. The earliest and still the most widely used solution is the traditional mechanical lock-and-key system. Owing to its simplicity, low cost, and reliability, the lock-and-key mechanism continues to be used in homes, offices, lockers, and storage facilities across the world. However, despite its widespread acceptance, a key-based system possesses a significant inherent weakness: the overall security of the vault is completely dependent on the physical key itself. If the key is misplaced, stolen, or secretly duplicated without the owner's knowledge, the security of the vault is immediately compromised. In many such situations, the owner remains completely unaware that unauthorized access is possible until after a theft or security breach has already occurred. This limitation makes conventional locking systems increasingly unsuitable for protecting high-value assets in today's security-conscious world.

To overcome the dependency on physical keys, modern locker systems have gradually shifted towards electronic keypad or password-based access mechanisms. By replacing the physical key with a user-defined passcode, these systems eliminate the possibility of losing keys or having them duplicated without permission. Password-based systems also provide greater convenience, as users only need to remember a secure code instead of carrying a key at all times. Furthermore, changing the access code is generally much easier than replacing an entire mechanical lock. Despite these improvements, most commercially available password-based lockers remain fundamentally passive security devices. Their primary function is limited to verifying whether an entered password matches the stored credentials. If the correct password is entered, access is granted; otherwise, access is denied. Beyond this basic functionality, they provide little or no awareness of what is happening around the vault. They are generally unable to detect physical tampering, forced entry attempts, vibration-based attacks, or unauthorized interference with the vault body. Moreover, they do not provide any means of informing the owner about suspicious activity, nor do they support remote monitoring or remote control when the owner is away from the premises. Consequently, while password-based systems improve convenience, they still leave several important security challenges unresolved.

With the rapid advancement of embedded systems, wireless communication technologies, and the Internet of Things (IoT), it has become possible to transform conventional lockers into intelligent security systems capable of monitoring, communication, and real-time response. IoT-enabled security solutions not only authenticate authorized users but also continuously observe the condition of the protected device, detect abnormal events, and communicate important information to the owner through cloud services. Such systems provide significantly higher levels of security, convenience, and situational awareness than conventional standalone lockers.

Motivated by these observations, this project was developed during a summer training program under the guidance of Mr. Armaan Hussain at Uptech Automation. The objective of the project was to bridge the gap between traditional password-based lockers and modern intelligent security systems by developing a practical prototype of an IoT and Embedded System Based Secure Vault. The prototype has been designed around the ESP32 microcontroller, selected for its powerful processing capability, integrated Wi-Fi connectivity, and suitability for embedded IoT applications. The system preserves the familiar keypad-based authentication method for convenient local access while introducing several advanced security features. These include an anti-tamper detection mechanism that continuously monitors the vault for forceful interference and immediately activates an audible alarm upon detecting suspicious vibrations. In addition, the system is integrated with the ESP RainMaker cloud platform, enabling remote lock and unlock functionality, cloud connectivity, and instant push notifications to alert the owner whenever a security event occurs. The prototype also maintains a live status display showing the most recent operational or security event, thereby providing users with immediate information regarding the condition of the vault. By combining embedded system design, cloud-based IoT technology, and real-time security monitoring, this project demonstrates an effective approach towards building a smarter, more secure, and user-friendly vault suitable for modern security requirements.

1.2 Project Analysis

1.2.1 About the Problem

Locker and vault security is an essential requirement in residential, commercial, and institutional environments where valuable assets such as cash, jewellery, confidential documents, and electronic storage devices must be protected against unauthorized access. Although numerous locking solutions are available today, the majority of security systems used in homes, retail shops, and small offices can still be broadly classified into two categories.

The first category consists of **traditional key-based lockers**, which rely entirely on a mechanical lock and a physical key for authentication. These systems are simple, cost-effective, and have been widely used for decades due to their reliability and ease of operation. However, their security depends completely on the possession of the physical key. If the key is lost, stolen, or secretly duplicated, the entire security mechanism becomes ineffective. More importantly, such unauthorized access can occur without leaving any indication that the vault has been compromised. As a result, the owner may remain completely unaware of the breach until valuable items are found to be missing, making the system reactive rather than preventive.

The second category includes **modern password- or keypad-based lockers**, which eliminate the dependence on a physical key by allowing users to authenticate using a numerical password or PIN. These systems offer improved convenience, as users no longer need to carry or safeguard a physical key, and passwords can be changed whenever required. While this represents a significant improvement over conventional mechanical locks, most commercially available keypad-based lockers continue to function as standalone access-control devices. Their primary responsibility is limited to verifying the correctness of the entered password and granting or denying access accordingly. They generally lack the ability to detect physical attacks such as prying, drilling, vibration, or forceful attempts to open the vault. Furthermore, they do not provide any mechanism for notifying the owner of suspicious activity or allowing the vault to be monitored and controlled remotely through the internet.

Consequently, neither of these categories adequately addresses two practical security challenges encountered in real-world applications. The first challenge is the **real-time detection of an active break-in attempt**, enabling the system to respond immediately rather than informing the owner only after a successful theft has occurred. The second challenge is the **lack of remote accessibility**, where owners have no means of checking the status of the vault, receiving instant security alerts, or controlling access when they are away from the premises. These limitations reduce the overall effectiveness of conventional locker systems in today's increasingly connected and security-conscious environment.

The core problem addressed in this project is therefore the design and implementation of an intelligent vault security system that combines the convenience of password-based authentication with the advanced capabilities of embedded systems and the Internet of Things (IoT). The proposed solution is intended to detect unauthorized physical tampering using vibration sensing, generate an immediate local alarm during suspicious activity, provide real-time notifications through cloud connectivity, and enable remote lock/unlock control via the ESP RainMaker platform. At the same time, the system is designed to distinguish between legitimate user operations and actual intrusion attempts, thereby minimizing false alarms during routine handling of the vault. By integrating these features into a single embedded platform, the project aims to provide a practical, affordable, and reliable smart security solution suitable for homes, shops, and small offices.

1.2.2 Why is the Particular Topic Chosen?

Locker and vault security is an essential requirement in residential, commercial, and institutional environments where valuable assets such as cash, jewellery, confidential documents, and electronic storage devices must be protected against unauthorized access. Although numerous locking solutions are available today, the majority of security systems used in homes, retail shops, and small offices can still be broadly classified into two categories.

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1.2.3 Objective and Scope of the Project

The primary objective of this project is to design, develop, and implement a functional prototype of an **IoT and Embedded System Based Secure Vault** that enhances the security offered by conventional password-based lockers through the integration of intelligent monitoring, tamper detection, and cloud connectivity. The system is intended to provide a secure, reliable, and user-friendly solution capable of protecting valuable assets while allowing both local and remote operation.

The first objective is to provide secure local authentication using a **4-digit keypad PIN**. Access to the vault should only be granted after successful password verification. To improve protection against unauthorized access, the system incorporates a security lockout mechanism that temporarily disables further password attempts after multiple consecutive incorrect entries, thereby reducing the possibility of brute-force password attacks.

A second objective is to enable **remote lock and unlock functionality** through the ESP RainMaker mobile application. To ensure that remote access cannot be misused, this feature is intentionally restricted to operate only when the user has explicitly selected the **Phone/RainMaker Access Mode** from the keypad menu. This additional layer of authorization prevents accidental or unauthorized cloud-based operation while maintaining the convenience of remote access whenever required.

Another important objective is to improve the physical security of the vault by incorporating an **anti-tamper detection system**. A vibration sensor continuously monitors the vault for abnormal impacts or forceful attempts to open it. The system is carefully calibrated to distinguish genuine tampering events from normal handling or minor vibrations, thereby minimizing false alarms. Whenever an intrusion attempt is detected, the vault immediately activates an audible buzzer alarm, displays an intrusion warning on the LCD, and sends a real-time push notification to the user's smartphone through the ESP RainMaker cloud platform.

The project also aims to provide clear and continuous **real-time system feedback** through a 16×2 LCD display. The display informs the user of important operational states such as *Ready*, *Enter Password*, *Wrong Password*, *Vault Locked*, *Vault Unlocked*, *Phone Access Enabled*, and *Intruder Alert*. This improves usability by allowing users to easily understand the current status of the system during operation.

Another key objective is to ensure the reliable operation of the embedded hardware. The system is designed to function without spurious microcontroller resets, brownout conditions, unstable power behaviour, or false sensor triggering. Particular attention has been given to hardware integration and power management to achieve stable performance on the fabricated printed circuit board (PCB).

The scope of this project includes the complete design and implementation of the embedded firmware using a **non-blocking state-machine architecture**, enabling responsive multitasking without the use of blocking delays. It also covers keypad interfacing, LCD communication, servo motor control for the locking mechanism, vibration sensor calibration, cloud integration through the ESP RainMaker IoT platform, mobile application connectivity, event notification, and hardware debugging. Furthermore, the project includes the design validation, PCB implementation, power-stability improvements, and system testing required to demonstrate a reliable, practical, and scalable smart vault suitable for applications in homes, offices, and small commercial establishments.

1.2.4 What Contribution Would the Project Make?

This project demonstrates a practical approach to enhancing the security and functionality of conventional password-based locker systems by integrating embedded systems and Internet of Things (IoT) technologies. While existing keypad-based lockers successfully eliminate the dependency on physical keys, they generally remain standalone systems with limited capabilities beyond basic password verification. The developed prototype addresses two major shortcomings commonly found in such systems: the absence of physical tamper detection and the inability to monitor or control the vault remotely. By incorporating these features into a single embedded platform, the project illustrates how traditional locker systems can be transformed into intelligent security devices without significantly increasing cost or system complexity.

A significant contribution of this work is the use of **low-cost, readily available hardware components**, including the ESP32 microcontroller, matrix keypad, vibration sensor, servo motor, LCD display, and buzzer, together with the **ESP RainMaker** cloud platform. The use of commercially available components makes the proposed system affordable, easy to reproduce, and suitable for educational purposes, prototype development, and small-scale real-world applications. Furthermore, leveraging an established IoT platform eliminates the need for developing a dedicated cloud infrastructure while still providing reliable remote connectivity, smartphone-based control, and real-time push notifications.

Another important contribution of this project is the documentation of a **practical embedded systems debugging process**. During hardware implementation, the prototype experienced brownout resets caused by the high transient current drawn by the servo motor during operation. The systematic identification of the root cause, followed by appropriate hardware modifications and power-stability improvements, demonstrates a realistic engineering problem-solving approach that is frequently encountered in embedded system design. The documented troubleshooting process can serve as a useful reference for students, hobbyists, and engineers developing similar embedded hardware projects.

Beyond demonstrating a functional prototype, this project establishes a flexible and scalable foundation for future research and product development. The modular architecture allows additional security features such as biometric fingerprint authentication, facial recognition using camera modules, RFID or NFC-based access control, encrypted event logging, cloud-based access history, battery backup, and AI-assisted intrusion detection to be incorporated with minimal modification to the existing design. Consequently, the project not only fulfills its current objectives but also provides a strong platform for the development of more advanced smart security systems in the future.

2. LITERATURE REVIEW

Electronic access control systems have become an important area of research and development due to the increasing demand for secure, reliable, and intelligent methods of protecting valuable assets. Conventional mechanical locks have gradually been supplemented and, in many cases, replaced by electronically controlled locking systems that offer greater convenience, flexibility, and security. Recent advances in embedded systems, wireless communication, and Internet of Things (IoT) technologies have further transformed electronic locking systems from standalone devices into connected security solutions capable of remote monitoring, event notification, and intelligent decision-making. Numerous studies and practical implementations have explored different approaches to electronic access control, tamper detection, cloud connectivity, and embedded hardware reliability. This review discusses the major categories of technologies relevant to the present project and highlights the research gap addressed by the proposed smart secure vault.

One of the earliest and most widely adopted forms of electronic access control is the **keypad-based locking system**. In these systems, users authenticate themselves by entering a predefined Personal Identification Number (PIN) through a matrix keypad. The entered password is compared with a stored value inside the microcontroller's memory, and access is granted only if the two match. The locking mechanism is typically operated using either a servo motor or a solenoid lock. Compared with traditional mechanical locks, keypad-based systems eliminate the need for carrying physical keys and remove the possibility of unauthorized key duplication. They also allow passwords to be changed easily without replacing the physical locking mechanism. Because of their simplicity and relatively low cost, keypad-based locking systems are widely used in homes, offices, lockers, hotels, laboratories, and educational institutions.

Despite these advantages, conventional keypad-based systems have significant limitations. Most available implementations focus exclusively on verifying user credentials while providing little awareness of the physical state of the locker. Once the password verification process is complete, the system generally performs no further monitoring of the vault. Consequently, if an attacker attempts to force open the locker by drilling, prying, or striking it with tools, the system remains completely unaware unless additional sensing mechanisms are incorporated. Furthermore, many low-cost keypad lockers do not maintain any event history, notify the owner of suspicious activity, or provide any remote access capability. These shortcomings limit their effectiveness in situations where continuous monitoring and timely response are important.

Another commonly used approach to electronic access control is the **Radio Frequency Identification (RFID)-based locking system**. RFID technology replaces passwords with contactless identification cards or tags containing unique identification numbers. When a valid RFID tag is brought close to the reader, the microcontroller compares the received identifier with stored records and unlocks the system if authorization is confirmed. RFID-based systems reduce the need to remember passwords and allow quick, convenient access. They have therefore found widespread application in attendance systems, hotels, office entry systems, libraries, warehouses, and industrial facilities.

However, RFID systems also possess several limitations. The security of the system depends entirely on the possession of the RFID tag. If the tag is lost, stolen, or cloned using specialized equipment, unauthorized users may gain access without needing to know any password. In addition, many basic RFID locker implementations operate as isolated embedded systems without communication capabilities. Similar to keypad-based lockers, they generally lack tamper detection, cloud connectivity, and real-time owner notification. Consequently, while RFID technology improves convenience, it does not completely address modern security requirements.

Researchers have also investigated **biometric authentication systems** for high-security applications. Biometric methods authenticate users using unique physiological characteristics such as fingerprints, facial features, iris patterns, or voice recognition. Since biometric identifiers are unique to individuals and difficult to duplicate, these systems generally offer higher security than passwords or RFID cards. Fingerprint-based lockers, in particular, have become increasingly popular because of the affordability and availability of fingerprint sensors compatible with embedded microcontrollers.

Although biometric systems significantly improve authentication security, they introduce additional challenges. Fingerprint sensors may experience reduced accuracy when fingers are dirty, wet, injured, or improperly positioned. Facial recognition systems require cameras with sufficient lighting conditions and greater computational resources. Moreover, biometric hardware increases the overall system cost and complexity. For many residential and educational applications, a well-designed keypad-based system enhanced with intelligent monitoring and cloud connectivity provides a more economical balance between cost, usability, and security.

Before cloud-connected IoT platforms became widely available, researchers commonly employed **GSM and SMS-based security systems** to provide remote alerts. In these systems, a GSM module connected to the microcontroller sends an SMS notification whenever an unauthorized access attempt or alarm condition is detected. Some implementations also support receiving SMS commands to remotely lock or unlock the system. These solutions represented one of the earliest forms of remote security monitoring and demonstrated the benefits of informing users immediately when suspicious activity occurred.

Despite their usefulness, GSM-based systems have several disadvantages. SMS delivery depends on mobile network availability and may experience delays during periods of poor network coverage. Sending messages may also incur recurring service charges depending on the communication provider. Additionally, SMS-based interfaces provide limited interaction compared to modern smartphone applications, offering little support for real-time device status, graphical interfaces, firmware updates, or continuous cloud synchronization. As IoT platforms became more mature, Wi-Fi-based cloud communication gradually replaced GSM modules in many embedded security applications.

The rapid growth of the **Internet of Things (IoT)** has significantly changed the design philosophy of embedded security systems. IoT enables embedded devices to communicate continuously with cloud servers, allowing users to monitor and control devices remotely using smartphones or web applications. Modern microcontrollers such as the ESP8266 and ESP32 include built-in Wi-Fi connectivity, making cloud integration practical without requiring additional communication hardware. This has encouraged the development of intelligent security systems capable of providing real-time monitoring, event notifications, remote configuration, and firmware updates.

Among the available IoT platforms, **ESP RainMaker** has gained considerable popularity for ESP32-based projects because it provides an integrated cloud infrastructure specifically designed for embedded devices. The platform offers predefined device abstractions such as switches, sensors, lights, and power controls, significantly reducing software development effort. It also provides secure communication, smartphone applications for Android and iOS, push notifications, Over-the-Air (OTA) firmware updates, scheduling functionality, and device management services. Instead of developing custom cloud servers and mobile applications, developers can focus on implementing application-specific embedded logic while relying on the platform to manage communication and cloud synchronization.

Several IoT-based smart locker projects reported in the literature combine cloud connectivity with password authentication or RFID access. These systems allow users to remotely monitor device status and control locking mechanisms through mobile applications. However, many implementations continue to emphasize remote access while paying comparatively little attention to physical intrusion detection. Consequently, unauthorized attempts involving forceful impacts or mechanical attacks may remain undetected unless additional sensing hardware is integrated into the system.

An important aspect of modern vault security is **tamper detection**. Researchers have explored various sensing technologies including vibration sensors, accelerometers, piezoelectric sensors, magnetic switches, tilt sensors, and force sensors for detecting unauthorized interference. Among these, vibration sensors such as the SW-420 are widely used in educational and prototype projects because they are inexpensive, simple to interface, and capable of detecting mechanical disturbances. They are frequently employed in applications such as ATM security, equipment protection, anti-theft systems, and industrial machine monitoring.

One challenge associated with vibration-based sensing is the occurrence of **false alarms**. Mechanical vibration sensors can briefly activate even when subjected to minor knocks, accidental contact, or environmental vibrations. If the embedded software responds immediately to every sensor transition, the system may generate frequent nuisance alarms that reduce user confidence. To overcome this limitation, researchers recommend analysing sensor activity over a short observation period rather than reacting to a single instantaneous reading. Time-window analysis, burst sampling, debounce filtering, and threshold-based decision algorithms have been shown to improve detection reliability while reducing unnecessary alarm activation.

The present project adopts this engineering principle by implementing **burst-sampling tamper detection**. Instead of treating a single sensor pulse as evidence of intrusion, the software samples the vibration sensor repeatedly over a defined interval and evaluates the overall disturbance level before deciding whether an intrusion attempt has occurred. This approach significantly improves reliability by distinguishing routine handling from sustained forceful impacts.

Another critical area discussed in embedded systems literature is **power-supply stability**. Although many prototype projects focus primarily on software functionality, practical embedded systems often encounter hardware issues arising from unstable power delivery. Servo motors, DC motors, relays, and similar electromechanical actuators draw large transient currents when starting or changing direction. These sudden current demands may produce voltage drops that activate the microcontroller's brownout detection circuitry, causing unexpected resets or unstable system behaviour.

Manufacturer application notes and embedded hardware design guidelines consistently recommend separating power supplies where practical, using adequately rated voltage regulators, providing sufficient current margins, and placing bulk decoupling capacitors close to high-current loads. Proper PCB layout, short power traces, and appropriate grounding techniques further improve system stability. During the implementation of this project, repeated brownout resets were observed whenever the servo motor operated. Systematic hardware debugging identified transient current spikes as the root cause. The issue was successfully resolved through power-supply improvements and the addition of suitable decoupling capacitance, demonstrating the importance of combining sound hardware design practices with embedded software development.

Another software engineering consideration frequently discussed in embedded systems research is the use of **non-blocking program architectures**. Early embedded projects often relied on long software delays using blocking functions, which prevented the processor from responding to other events during execution. Modern embedded systems increasingly employ finite state machines, event-driven programming, and timer-based scheduling to improve responsiveness. Such architectures enable multiple tasks—including keypad scanning, LCD updates, sensor monitoring, cloud communication, and alarm handling—to execute concurrently without reducing system responsiveness. The present project follows this approach by implementing the firmware using a non-blocking state-machine architecture, thereby ensuring smooth and reliable operation of all subsystems.

Overall, the reviewed literature demonstrates that the individual technologies used in this project—including keypad authentication, IoT cloud connectivity, vibration-based intrusion detection, embedded firmware design, and power-supply stabilization—have each been extensively investigated. However, many published educational prototypes focus on only one or two of these aspects. Systems emphasizing remote IoT control often omit reliable tamper detection, while tamper detection projects frequently lack cloud integration or robust software architecture. Similarly, relatively few prototype implementations document practical hardware debugging challenges encountered during real PCB development.

The present project addresses this gap by integrating these well-established technologies into a single, coherent embedded security system. It combines secure keypad authentication, access-mode-restricted remote control through ESP RainMaker, false-alarm-resistant vibration-based tamper detection, real-time LCD status indication, smartphone push notifications, and reliable embedded hardware operation into one functional prototype. In addition to demonstrating the successful integration of these technologies, the project documents the practical engineering decisions, calibration methods, and hardware reliability improvements required to transform individual concepts into a dependable smart vault system suitable for homes, offices, educational laboratories, and small commercial applications.

3. SYSTEM DESIGN

The system design of the IoT and Embedded System Based Secure Vault is centred around the ESP32 microcontroller, which coordinates all sensors, actuators, the display, and the cloud connection through a single non-blocking state machine. The firmware architecture ensures that safety-critical checks — the vibration/tamper sensor and incoming RainMaker commands — are evaluated on every pass of the control loop, regardless of which screen or access mode is currently displayed.

3.1 Component Explanation

Component	Specification	Function
ESP32 Development Board	Dual-core, Wi-Fi + BLE	Central controller running the state machine and RainMaker connection
4x4 Matrix Keypad	16-key membrane keypad	PIN entry and menu navigation
16x2 LCD Display	HD44780 with I2C backpack	Real-time status display
Servo Motor	SG90-type, 60–90° travel	Physical vault locking mechanism
Ultrasonic Sensor	HC-SR04, ≈ 2–400 cm range	User presence detection
Vibration Sensor	SW-420 digital output	Anti-tamper / intrusion detection
Buzzer	5V active buzzer	Audible alerts and alarm
Decoupling Capacitor	3300µF, 16V electrolytic	Power rail stabilisation for the servo

Table 3.1: Hardware Components and Their Function

1. ESP32 Development Board

- Acts as the central controller of the entire vault, running the non-blocking state machine that manages the keypad, LCD, servo, ultrasonic sensor, vibration sensor, and the ESP RainMaker cloud connection.
- Provides built-in Wi-Fi and Bluetooth Low Energy (BLE), used respectively for the RainMaker cloud connection and for initial device provisioning.

Fig 3.1: ESP32 Development Board

2. 4x4 Matrix Keypad

- Used for entering the 4-digit PIN and for navigating the on-screen menu (Phone Mode / Keypad Mode).
- Keys 'A' through 'D' are used for menu selection, cancel, and lock confirmation, in addition to the numeric keys for the PIN.

Fig 3.2: 4x4 Matrix Keypad

3. 16x2 I2C LCD Display

- Displays real-time system status — Ready, Welcome, Password entry, Access Granted, Wrong Password, Lockout Countdown, and Intruder Alert.
- Communicates over the I2C bus, minimising the number of GPIO pins required.

Fig 3.3: 16x2 I2C LCD Display

4. Servo Motor (Locking Mechanism)

- Drives the physical locking bolt between a locked position (60°) and an unlocked position (90°).

- Identified during testing as the source of a current-spike-induced brownout reset, resolved with a decoupling capacitor (see Section 3.1, Item 8).

Fig 3.4: SG90 Servo Motor

5. HC-SR04 Ultrasonic Sensor

- Detects the presence of a user approaching the vault (within approximately 50 cm) and triggers the welcome screen and access menu.

Fig 3.5: HC-SR04 Ultrasonic Sensor

6. SW-420 Vibration Sensor

- Forms the core of the anti-tamper system, detecting physical disturbance to the vault enclosure.
- Rather than reacting to a single instantaneous reading, the firmware takes a short rapid-sampling burst (approximately 120 ms) the moment a disturbance is first detected, and counts how many samples within that burst indicate a triggered state. A light incidental touch produces only a few triggered samples and is ignored; a genuine hard hit keeps the sensor's internal switch bouncing for longer, producing many triggered samples in the same short burst, and is classified as a tamper event.
- The alarm, once triggered, remains active — buzzer sounding and red LED blinking rapidly — until cleared by an authorised keypad entry ('D') or a remote silence command from the RainMaker app.

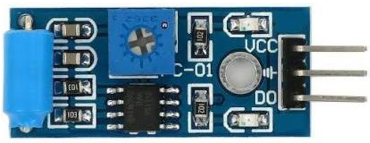

Fig 3.6: SW-420 Vibration Sensor

7. Buzzer Module

- Provides audible feedback for a wrong password entry and for the intruder alarm.

Fig 3.7: Buzzer Module

8. 3300µF / 16V Electrolytic Capacitor

- Added across the main power rail to resolve a recurring brownout reset that occurred at the instant the servo motor began moving. The servo's current spike caused a momentary voltage dip large enough to trip the ESP32's brownout detector, resetting the entire system mid-operation.
- The capacitor acts as a local charge reservoir, supplying the instantaneous current demand of the servo without allowing the main supply rail to sag, thereby preventing the brownout reset.

Fig 3.8: 3300µF / 16V Electrolytic Capacitor

Component / Signal	ESP32 GPIO Pin	Mode
Ultrasonic Trigger (TRIG_PIN)	GPIO 19	Output
Ultrasonic Echo (ECHO_PIN)	GPIO 34	Input
Servo Signal (SERVO_PIN)	GPIO 5	PWM Output
Green LED	GPIO 13	Output
Red LED	GPIO 12	Output
Buzzer	GPIO 18	Output
Vibration Sensor (SW-420)	GPIO 16	Input
LCD – SDA	GPIO 21	I2C
LCD – SCL	GPIO 22	I2C
Wi-Fi / Factory Reset Button	GPIO 0 (BOOT)	Input

Table 3.2: ESP32 GPIO Pin Configuration

3.2 Software Flow

The firmware of the proposed secure vault is implemented entirely using a **non-blocking finite state machine (FSM) architecture**, which enables the system to perform multiple tasks efficiently without interrupting its normal operation. Unlike conventional embedded programs that rely on long blocking delays or continuous while(true) loops, the proposed firmware allows all system functions to execute concurrently by repeatedly cycling through the main loop() function. This design ensures that the microcontroller remains responsive to user inputs, cloud commands, and security events at all times.

Each operational mode of the vault is represented as an independent state within the finite state machine. These states include system initialization, idle monitoring, welcome message display, menu selection, keypad password entry, phone/RainMaker access mode, password verification, vault locking and unlocking, lockout countdown after multiple incorrect password attempts, intruder alarm handling, and status display. Every state contains only the logic necessary for its specific operation and determines the conditions under which the system transitions to another state. This modular organization improves code readability, simplifies debugging, and makes future feature additions easier without affecting the overall program structure.

During every iteration of the Arduino loop() function, the firmware follows a fixed execution sequence. It first checks for incoming commands from the ESP RainMaker cloud platform to determine whether a remote lock or unlock request has been received. Immediately afterward, it continuously monitors the vibration sensor for any indication of physical tampering. Since these critical checks are performed on every execution cycle, the system is capable of responding almost instantaneously to both remote user commands and intrusion attempts, irrespective of the current operating state.

After processing these high-priority events, the firmware executes the logic associated with the current state and performs only a small portion of the required task before returning to the beginning of the loop() function. Because no blocking delays are used, other important activities such as keypad scanning, LCD updates, servo motor control, countdown timers, and cloud communication continue to operate smoothly and simultaneously. This prevents the system from becoming unresponsive while waiting for user input or during time-dependent operations.

The non-blocking state-machine approach significantly improves the overall reliability and responsiveness of the secure vault. It ensures that tamper events, remote commands, keypad inputs, and display updates are handled without delay, eliminating the possibility of missed events that could occur in a blocking implementation. Furthermore, the modular firmware architecture provides a scalable foundation for incorporating additional features such as biometric authentication, access logging, multiple user accounts, or advanced intrusion detection algorithms in future versions of the system with minimal modification to the existing codebase.

3.3 Mode-wise Functional Flow

1. Idle / Detection Mode

The system normally remains in the **Idle Mode**, which serves as the default operating state after power-up or after completing a previous operation. During this state, the ultrasonic sensor continuously measures the distance in front of the vault to detect the presence of a user. Since the firmware follows a non-blocking architecture, distance measurements are performed periodically without interrupting other background activities such as cloud communication and vibration monitoring.

When the ultrasonic sensor detects a person within the predefined threshold distance, the system assumes that a user intends to operate the vault. It then displays a welcome message on the LCD screen for a short duration before automatically transitioning to the **Access Mode Selection Menu**. If no user is detected, the system continues monitoring while remaining in the idle state.

2. Menu Selection

After detecting a user, the vault presents an access menu on the LCD, allowing the user to choose between two authentication methods. The selection is performed using the keypad.

- **Option A – Phone / ESP RainMaker Mode**
- **Option B – Keypad Password Mode**

The selected option determines the subsequent operating state. By requiring the user to explicitly choose the desired authentication method, the system prevents unintended operations and provides an additional layer of security before any unlocking action is permitted.

3. Phone / RainMaker Mode

When **Phone Mode** is selected, the vault enters a waiting state in which it continuously listens for commands received through the ESP RainMaker cloud platform. During this period, the user can remotely lock or unlock the vault using the ESP RainMaker mobile application.

A key security feature of the system is that **remote unlock commands are accepted only while the vault is actively waiting in Phone Mode**. If a cloud command is received while the vault is in any other operating state—such as Idle Mode, Password Entry Mode, Lockout Mode, or Intruder Alarm Mode—it is intentionally ignored. This design prevents accidental or unauthorized remote unlocking when the owner has not explicitly requested remote access through the keypad interface.

The LCD continuously displays that Phone Mode is active, informing the user that the system is awaiting a cloud command.

4. Keypad Password Mode

If the user selects **Keypad Password Mode**, the LCD prompts for a four-digit Personal Identification Number (PIN). As each key is pressed, the firmware validates the input and stores the entered digits until the complete password is received.

If the entered PIN matches the stored password, the servo motor rotates to unlock the vault and the LCD displays a confirmation message indicating successful authentication. After a predefined delay, the vault automatically returns to its secure locked state.

If an incorrect password is entered, the system increments the wrong-attempt counter, activates the buzzer briefly, and displays the number of remaining attempts on the LCD. This immediate feedback informs the user without revealing the correct password.

To protect against brute-force password attacks, the firmware implements a security lockout mechanism. After **three consecutive incorrect password attempts**, the vault disables password entry for approximately **30 seconds**. During this period, the LCD displays a countdown timer indicating the remaining lockout duration. Once the timer expires, the wrong-attempt counter is automatically reset and the user is allowed to attempt authentication again.

5. Intruder / Tamper Alarm Mode

The **Intruder Alarm Mode** is activated whenever the burst-sampling vibration detection algorithm identifies sustained forceful impacts on the vault. Unlike simple threshold-based detection, the implemented algorithm evaluates multiple sensor readings over a short observation window before declaring an intrusion event, thereby significantly reducing false alarms caused by accidental touches or environmental vibrations.

Upon detecting a genuine tamper attempt, the firmware immediately performs several security actions simultaneously. The buzzer generates a continuous alarm sound, the red LED flashes rapidly to provide a clear visual warning, the LCD displays an **"Intruder Alert"** message, and an instant push notification is transmitted to the user's smartphone through the ESP RainMaker cloud platform.

The alarm continues until it is manually cleared either by pressing the designated **'D' key** on the keypad or through the remote alarm-silence command available in the ESP RainMaker application. After the alarm has been cleared, the firmware activates a short cooldown period before resuming normal monitoring. This prevents repeated triggering from residual vibrations immediately following the original disturbance and improves the stability of the intrusion detection system.

3.4 Initial System Architecture

The initial implementation of the secure vault focused primarily on achieving basic password-protected access using a matrix keypad, servo motor-based locking mechanism, and LCD-based user interface. At this stage, the system successfully authenticated users through a predefined password and controlled the locking mechanism accordingly. Basic menu navigation and servo actuation were functional, providing a proof of concept for the overall access-control mechanism.

However, the early firmware architecture relied heavily on **blocking while(true) loops** to manage different operational modes. Each major function—such as password entry, menu handling, and waiting for user interaction—executed inside an independent infinite loop that prevented the processor from performing other tasks until that operation was completed.

Although this programming approach simplified the initial implementation, it introduced several significant limitations. While the microcontroller was occupied inside one operational loop, it could not continuously monitor the vibration sensor for intrusion attempts or process remote commands received through the ESP RainMaker cloud platform. As a result, an intruder event occurring during password entry or menu navigation could remain undetected, and remote lock or unlock commands might not be processed until the current loop had finished executing.

These shortcomings motivated a complete redesign of the firmware architecture. The blocking implementation was replaced with a **non-blocking finite state machine**, allowing every subsystem—including keypad scanning, LCD updates, ultrasonic sensing, vibration monitoring, servo control, timers, and cloud communication—to operate concurrently within the Arduino loop() function. This architectural change greatly improved system responsiveness, ensured that no critical events were missed, and provided a robust foundation for integrating advanced IoT security features into the final prototype.

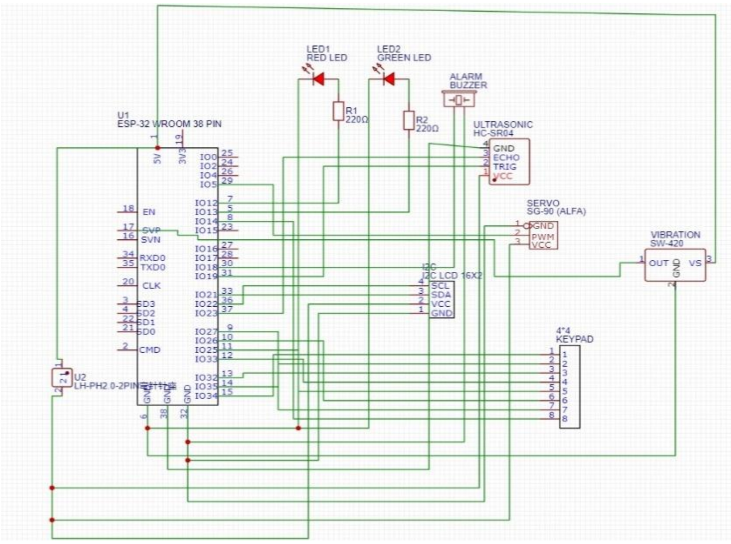

Fig 3.9: Initial System Block Diagram

3.5 Final Enhanced System Architecture

The final version of the secure vault represents a significant improvement over the initial prototype in terms of functionality, reliability, and overall system architecture. Based on the limitations identified during the early stages of development, the firmware was completely redesigned using a **non-blocking finite state machine (FSM)** architecture. This approach eliminated the dependency on blocking while(true) loops and enabled the microcontroller to execute multiple tasks efficiently within the Arduino loop() function. As a result, the system became capable of continuously monitoring sensors, processing keypad inputs, communicating with the ESP RainMaker cloud platform, and updating the LCD simultaneously without compromising responsiveness.

One of the major enhancements introduced in the final version is the implementation of a **burst-sampling anti-tamper detection algorithm**. Instead of activating the alarm immediately after detecting a single vibration pulse, the firmware samples the vibration sensor repeatedly over a short observation interval and evaluates the overall disturbance level before determining whether a genuine intrusion attempt has occurred. This approach significantly reduces false alarms caused by accidental touches, minor vibrations, or normal handling of the vault while ensuring reliable detection of forceful tampering.

To further improve alarm stability, a **post-alarm re-arm cooldown mechanism** was incorporated into the firmware. Once the alarm has been acknowledged and cleared, the system temporarily suspends tamper detection for a brief predefined period. This prevents repeated alarm activation due to residual vibrations immediately following the original disturbance, thereby improving user experience and preventing unnecessary repeated notifications.

The visual indication of security events was also enhanced through the addition of **rapid red LED blinking** during an active intrusion alarm. While the buzzer provides an audible warning, the flashing red LED serves as a clear visual indication of unauthorized activity, making the alarm more noticeable in noisy environments or situations where the buzzer may not be easily heard.

Another important security improvement is the implementation of **access-mode-restricted remote unlocking**. In the final design, commands received from the ESP RainMaker cloud platform are not accepted unconditionally. Remote lock and unlock operations are permitted only when the user has explicitly selected the **Phone/RainMaker Access Mode** from the keypad menu. Any cloud command received while the vault is operating in other states, such as Idle Mode, Password Entry Mode, Lockout Mode, or Intruder Alarm Mode, is intentionally ignored. This additional authorization layer prevents unintended or unauthorized remote unlocking and strengthens the overall security of the system.

The final hardware design also addressed a critical **power-supply stability issue** encountered during testing. Initial experiments revealed that the high instantaneous current drawn by the servo motor during operation caused temporary voltage drops, triggering the ESP32's brownout protection and resulting in unexpected system resets. To eliminate this problem, a **bulk decoupling capacitor** was connected close to the servo motor's power supply. The capacitor supplies the transient current required during servo startup, reducing voltage fluctuations and ensuring stable operation of the microcontroller. Following this modification, the system operated reliably without brownout resets, even during repeated locking and unlocking operations.

Overall, these improvements transformed the initial proof-of-concept into a robust and dependable smart vault system. The combination of a non-blocking firmware architecture, intelligent tamper detection, controlled cloud-based access, enhanced alarm mechanisms, and improved hardware reliability resulted in a secure, responsive, and scalable embedded system suitable for practical IoT-based security applications.

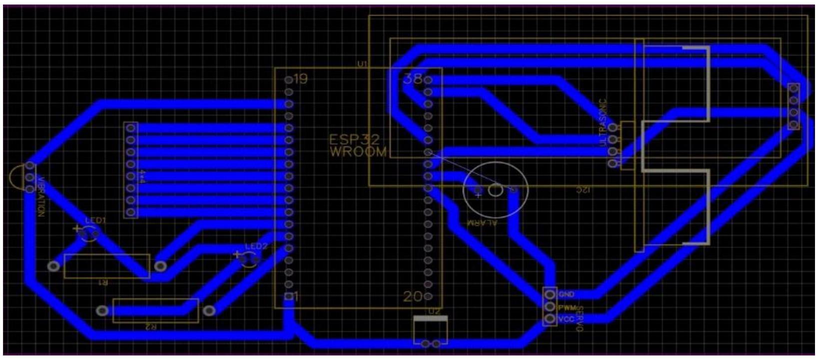

Fig 3.10: Final Enhanced System Block Diagram

4. RESULT AND DISCUSSION

The completed prototype was tested across all of its intended functional scenarios, including keypad-based authentication, wrong-password lockout, ESP RainMaker phone-based access, and vibration-based tamper detection. The system was also evaluated for hardware stability during repeated lock and unlock cycles to verify reliable operation under continuous use. All functional modules operated as expected, with correct LCD status updates, servo motor actuation, buzzer and LED indications, and cloud-based notifications. The implemented power-supply improvements successfully eliminated brownout resets, while the burst-sampling tamper detection algorithm minimized false alarms and reliably detected genuine intrusion attempts. Overall, the prototype demonstrated stable, responsive, and dependable performance throughout the testing process.

4.1 Functional Testing Scenarios and Observations

Test Scenario	Observation
Correct PIN entry	Vault unlocks smoothly, LCD shows “Access Granted”, green LED turns on, and a RainMaker push notification is sent.
Incorrect PIN entry	Buzzer sounds, LCD shows remaining attempts; after 3 incorrect attempts a 30-second lockout countdown is enforced before further attempts are allowed.
RainMaker unlock – inside Phone Mode	Vault unlocks immediately upon receiving the app command, as intended.
RainMaker unlock – outside Phone Mode	Command is correctly ignored; vault remains locked unless the user has explicitly selected Phone Mode from the menu.
Light incidental touch on vault	Ignored by the burst-sampling tamper logic; no false alarm triggered.
Genuine hard tamper hit	Alarm triggers immediately — buzzer sounds continuously, red LED blinks rapidly, and a push notification is sent.
Alarm silencing (keypad 'D' / app command)	Buzzer and red LED turn off immediately; a short re-arm cooldown prevents an instant false re-trigger.
Repeated unlock/lock cycling (pre-capacitor)	Intermittent brownout resets observed — LCD re-initialising, servo reverting to locked position, buzzer left stuck on.
Repeated unlock/lock cycling (post-capacitor)	No brownout resets observed; servo, buzzer, and LCD operate reliably across repeated cycles.

Table 4.1: Functional Testing Scenarios and Observations

4.2 Pictures of Project

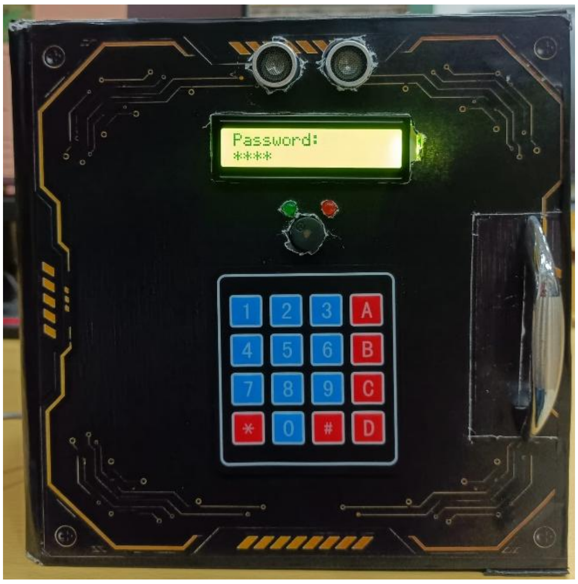

Fig 4.1: Completed Prototype – Front View

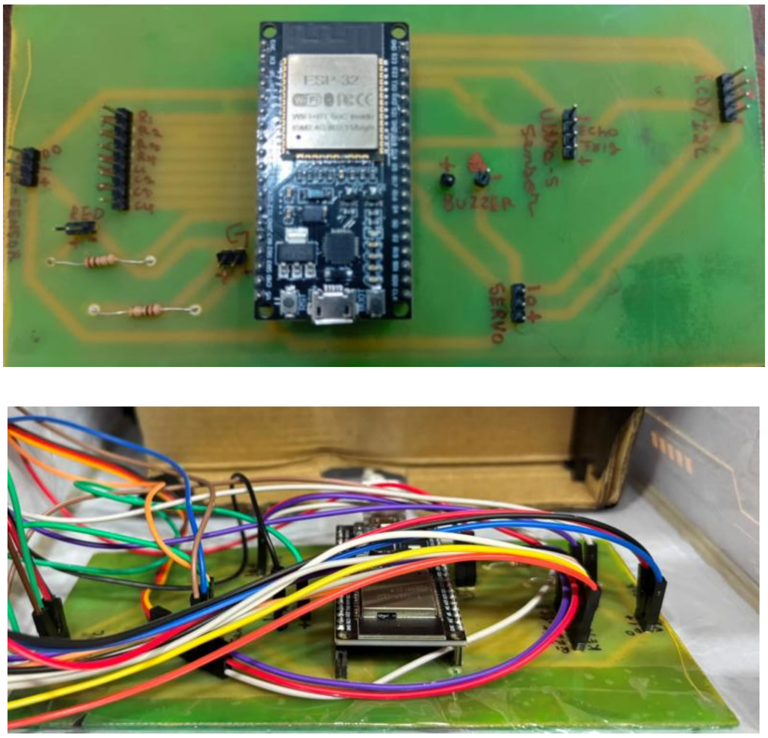

Fig 4.2: Completed Prototype – Internal Wiring

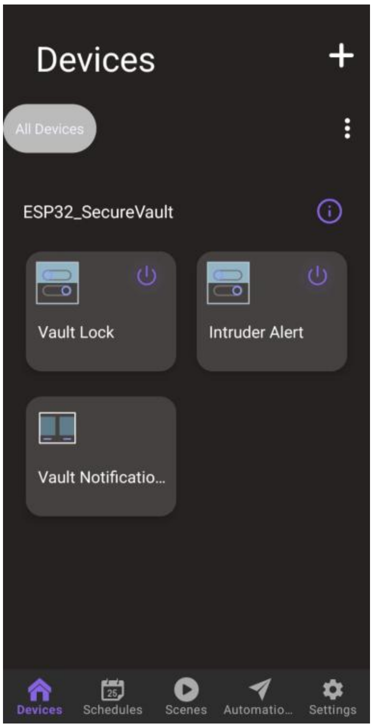

Fig 4.3: ESP RainMaker Mobile App Interface

4.3 Project Progress and Status

4.3.1 Work Completed

- Successfully designed and implemented the complete firmware using a **non-blocking finite state machine (FSM)** architecture. The earlier implementation based on blocking while(true) loops was replaced with a state-driven approach, enabling continuous monitoring of sensors, cloud communication, keypad input, LCD updates, and servo control without interrupting the execution of other tasks. This significantly improved the responsiveness and reliability of the overall system.

- Implemented and thoroughly tested a **4-digit keypad-based PIN authentication system** for secure local access. The password verification mechanism correctly authenticated valid users while denying unauthorized access. A built-in security feature was added to temporarily lock the system after three consecutive incorrect password attempts, thereby providing protection against brute-force password guessing attacks.

- Successfully integrated the prototype with the **ESP RainMaker IoT cloud platform**, enabling seamless communication between the embedded system and the smartphone application. The integration supports remote lock and unlock control, remote alarm-silence functionality, real-time push notifications for intrusion events, and a live **"Last Event"** status display that informs the user about the most recent activity performed on the vault.

- Improved the overall security of the system by implementing **access-mode-restricted remote unlocking**. Remote unlock commands are accepted only when the user has intentionally selected the **Phone/RainMaker Access Mode** through the keypad interface. Commands received while the vault is operating in any other mode are ignored, preventing unintended or unauthorized cloud-based unlocking.

- Redesigned the **anti-tamper detection mechanism** by replacing a simple threshold-triggering approach with a **burst-sampling vibration detection algorithm**. Instead of responding to a single sensor pulse, the firmware analyses multiple vibration samples collected over a short time interval before declaring an intrusion event. This approach significantly reduced false alarms caused by minor touches, accidental handling, or environmental vibrations while maintaining reliable detection of genuine forceful impacts.

- Enhanced the intrusion alert system by incorporating **rapid red LED blinking** synchronized with the audible buzzer alarm. During a detected tamper event, both visual and audible indicators operate simultaneously to provide a clear warning. The alarm can be acknowledged and cleared either locally using the keypad or remotely through the ESP RainMaker application, after which the system enters a short cooldown period before resuming normal monitoring.

- Successfully diagnosed and resolved a recurring **ESP32 brownout reset** issue encountered during servo motor operation. Detailed hardware testing revealed that the high instantaneous current drawn by the servo motor caused temporary voltage drops on the power supply, triggering the ESP32 brownout protection circuit. The issue was eliminated by adding a **3300 µF, 16 V decoupling capacitor** across the main power rail, providing sufficient transient current and ensuring stable operation during repeated locking and unlocking cycles.

- Conducted comprehensive testing of all functional modules, including keypad authentication, menu navigation, remote cloud control, vibration-based intrusion detection, LCD status display, servo operation, buzzer and LED indications, and push notifications. All modules functioned correctly under normal operating conditions as well as during simulated intrusion scenarios.

- Verified the long-term reliability of the prototype through repeated operational testing involving multiple lock and unlock cycles. The system maintained stable performance without unexpected resets, communication failures, missed cloud commands, or sensor malfunctions, demonstrating the effectiveness of both the embedded firmware architecture and the hardware improvements.

- Overall, the developed prototype successfully achieved the objectives defined at the beginning of the project by combining secure password-based authentication, intelligent tamper detection, IoT-based remote monitoring and control, and reliable embedded hardware into a single integrated smart vault system. The project demonstrates a practical and scalable approach for developing affordable IoT-enabled security solutions suitable for homes, offices, laboratories, and small commercial applications.

4.3.2 Potential Future Enhancements

- **Implementation of an Access History Log:** Future versions of the system can maintain a detailed, time-stamped record of all important events, including successful unlocks, incorrect password attempts, remote access operations, and tamper alerts.

This log can be synchronized with the ESP RainMaker cloud platform and viewed through the mobile application or a dedicated dashboard, allowing users to monitor vault activity and review past security events whenever required.

- **Integration of a Camera Module:** A camera module such as the ESP32-CAM can be integrated into the system to capture images automatically whenever a tamper event or repeated failed authentication attempt is detected. The captured images can be stored locally or uploaded to cloud storage, providing visual evidence of unauthorized access attempts and improving the overall effectiveness of the security system.
- **Support for Multiple User Accounts:** The current implementation supports a single user password. Future enhancements can allow multiple users to have individual PINs or authentication credentials. Each user can be assigned a unique identity, enabling the system to maintain user-specific access records and improving accountability in environments where multiple authorized users share the same vault.
- **Battery Backup and Power Failure Protection:** A rechargeable battery backup system can be incorporated to ensure uninterrupted operation during mains power failures. The system can automatically switch to battery power while continuing to monitor for tamper events, process user authentication, and maintain cloud connectivity. Additionally, low-battery detection and notification features can alert the user through the ESP RainMaker application, ensuring timely battery replacement or recharging before complete power loss occurs.
- **Biometric Authentication:** The security of the vault can be further enhanced by integrating fingerprint or facial recognition modules. Combining biometric verification with keypad authentication would provide multi-factor authentication, making unauthorized access significantly more difficult.
- **Advanced Sensor Integration:** Future versions may incorporate additional sensors, such as magnetic reed switches, accelerometers, or force sensors, to improve intrusion detection accuracy. Sensor fusion techniques can further reduce false alarms while increasing the reliability of tamper detection.
- **Cloud-Based Analytics and Data Logging:** Future implementations can include long-term cloud storage of operational data, allowing users to analyse access patterns, generate security reports, and receive intelligent insights regarding unusual activities. Such features would make the system more suitable for commercial and industrial security applications.
- **Mobile Application Enhancements:** The smartphone interface can be extended with additional features such as user management, customizable notification settings, remote password updates, scheduling of access permissions, and geofencing-based security controls, thereby providing greater flexibility and convenience to users.

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